

Note that the $\dot{K}$ equation can be simplified to the following if Q , C , and Φ are constant:

$$\dot{K} = \dot{P}C^T \tilde{R}^{-1} \quad (8.127)$$

5. Compute the state estimate as

$$\hat{x} = z + Ky \quad (8.128)$$

■ EXAMPLE 8.4

Suppose that it is known that a continuous-time measurement $v(t)$ has a total power of 1 watt and a power spectrum that is bandlimited to frequencies below 10 Hz. In this example, we will use our knowledge of the frequency content of $v(t)$ to obtain a dynamic model for $v(t)$. The power spectrum $S_v(\omega)$ can be plotted as shown in Figure 8.1. The magnitude of the spectrum, $1/40\pi$, is obtained by realizing that the total power of the signal (1 watt) is equal to the integral from $-\infty$ to $+\infty$ of $S_v(\omega)$, and $S_v(\omega)$ is an even function of ω . The spectrum shown in Figure 8.1 can be approximated as

$$\begin{aligned} S_v(\omega) &\approx \frac{1/2}{\omega^2 + (20\pi)^2} \\ &= \left(\frac{1}{j\omega + 20\pi} \right) \left(\frac{1}{-j\omega + 20\pi} \right) \left(\frac{1}{2} \right) \\ &= G(\omega)G(-\omega)S_\phi(\omega) \end{aligned} \quad (8.129)$$

This shows that $v(t)$ is the output of a linear system with a transfer function of $G(\omega)$ and an input of $\phi(t)$, where $\phi(t)$ is white noise with a variance of $1/2$ (see Equation 3.75). This can be written in the s -domain and then translated to the time domain as follows:

$$\begin{aligned} V(s) &= G(s)\Phi(s) \\ &= \frac{\Phi(s)}{s + 20\pi} \\ sV(s) + 20\pi V(s) &= \Phi(s) \\ sV(s) &= -20\pi V(s) + \Phi(s) \\ \dot{v} &= -20\pi v + \phi \end{aligned} \quad (8.130)$$

where $\phi(t)$ is white noise with variance $\Phi = 1/2$. Additional discussion and examples of this method can be found in [Bur99].

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8.5 THE STEADY-STATE CONTINUOUS-TIME KALMAN FILTER

In some situations, the Kalman filter converges to an LTI filter. If this is the case then we can often get good filtering performance by using a constant Kalman gain K in the filter. Then we do not have to worry about integrating the differential Riccati equation to solve for P and we do not have to worry about updating K in

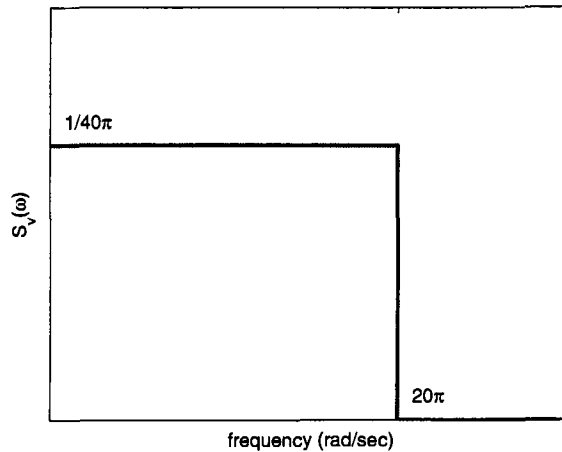


Figure 8.1 Power spectrum of bandlimited measurement noise for Example 8.4.

real time. This can provide a large savings in filter complexity and computational effort at the cost of only a small sacrifice of performance. In this section, we discuss the conditions under which the continuous-time filter converges to an LTI filter, and the steady-state filter's relationship to Wiener filtering and optimal control.

8.5.1 The algebraic Riccati equation

Recall from Equation (8.37) that the differential Riccati equation is given as

$$\dot{P} = -PC^TR^{-1}CP + AP + PA^T + Q \quad (8.131)$$

If A , C , Q , and R are constant (i.e., the system and measurement equations form an LTI system with constant noise covariances) then P may reach a steady-state value and $\dot{P}$ may eventually reach zero. This implies that

$$-PC^TR^{-1}CP + AP + PA^T + Q = 0 \quad (8.132)$$

This is called an algebraic Riccati equation (ARE). To be more specific, it is called a continuous ARE (CARE).³

The ARE solution may not always exist, and even if it does exist it may not result in a stable Kalman filter. We will summarize the most important Riccati equation convergence results below, but first we need to define what it means for a system to be controllable on the imaginary axis.

Definition 12 *The matrix pair (A, B) is controllable on the imaginary axis if there exists some matrix K such that $(A - BK)$ does not have any eigenvalues on the imaginary axis.*

³In the MATLAB Control System Toolbox the CARE can be solved by invoking the command $P = \text{CARE}(A^T, C^T, Q, R)$. The reason that the transposes are required is that MATLAB's CARE command is designed to solve the ARE for continuous-time optimal control problems. When we use it to solve for the Kalman filtering problem we need to transpose the A and C matrices, as discussed in Section 8.5.3.

This is similar to the concept of controllability on the unit circle for discrete-time systems (see Section 7.3). Now we summarize the most important Riccati equation convergence results from [Kai00], where proofs are given. Recall that the ARE is given as

$$-PC^TR^{-1}CP + AP + PA^T + Q = 0 \quad (8.133)$$

We assume that $Q \geq 0$ and $R > 0$. We define G as any matrix such that $GG^T = Q$. The corresponding steady-state Kalman gain K is given as

$$K = PC^TR^{-1} \quad (8.134)$$

The steady-state Kalman filter is given as

$$\dot{\hat{x}} = (A - KC)\hat{x} + Ky \quad (8.135)$$

We say that the CARE solution P is stabilizing if it results in a stable steady-state filter. That is, P is defined as a stabilizing CARE solution if all of the eigenvalues of $(A - KC)$ have negative real parts.

Theorem 27 *The CARE has a unique positive semidefinite solution P if and only if both of the following conditions hold.*

1. (A, C) is detectable.
2. (A, G) is stabilizable.

Furthermore, the corresponding steady-state Kalman filter is stable. That is, the eigenvalues of $(A - KC)$ have negative real parts.

This theorem is analogous to Theorem 23 for discrete-time Kalman filters. The above theorem does not preclude the existence of CARE solutions that are negative definite or indefinite. If such solutions exist, then they would result in an unstable Kalman filter. If we weaken the stabilizability condition in the above theorem, we obtain the following.

Theorem 28 *The CARE has at least one positive semidefinite solution P if and only if both of the following conditions hold.*

1. (A, C) is detectable.
2. (A, G) is controllable on the imaginary axis.

Furthermore, exactly one of the positive semidefinite ARE solutions results in a stable steady-state Kalman filter.

This theorem is analogous to Theorem 24 for discrete-time Kalman filters. This theorem states conditions for the existence of exactly one stabilizing positive definite CARE solution. However, there may be additional CARE solutions (positive definite or otherwise) that result in unstable Kalman filters. If a time-varying Kalman filter is run in this situation, then the Kalman filter equations may converge to either a stable or an unstable filter, depending on the initial condition $P(0)$. If we strengthen the controllability condition of Theorem 28, we obtain the following.

Theorem 29 *The CARE has at least one positive definite solution P if and only if both of the following conditions hold.*

1. (A, C) is detectable.
2. (A, G) is controllable in the closed left half plane.

Furthermore, exactly one of the positive definite CARE solutions results in a stable steady-state Kalman filter.

This theorem is analogous to Theorem 25 for discrete-time Kalman filters. If we drop the controllability condition in the above two theorems, we obtain the following.

Theorem 30 *The CARE has at least one positive semidefinite solution P if (A, C) is detectable. Furthermore, at least one such solution results in a marginally stable steady-state Kalman filter.*

This theorem is analogous to Theorem 26 for discrete-time Kalman filters. Note that the resulting filter is only marginally stable, so it may have eigenvalues on the imaginary axis. Also note that this theorem poses a sufficient (not necessary) condition. That is, there may be a stable steady-state Kalman filter even if the conditions of the above theorem do not hold. Furthermore, even if the conditions of the theorem do hold, there may be CARE solutions that result in unstable Kalman filters.

Additional results related to the stability of the steady-state continuous-time filter can be found many places, including [Aok67, Buc67, Buc68, Kwa72]. Many practical Kalman filters are applied to systems that do not meet the conditions of the above theorems, but the filters still work well in practice.

■ EXAMPLE 8.5

In this example we consider the following two-state system that is taken from [Buc68, Chapter 5]:

$$\begin{aligned} \dot{x} &= \begin{bmatrix} a_1 & 0 \\ 0 & a_2 \end{bmatrix} x + w \\ y &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} x + v \\ Q &= \begin{bmatrix} q_{11} & q_{12} \\ q_{12} & q_{22} \end{bmatrix} \\ R &= \begin{bmatrix} r_1 & 0 \\ 0 & r_2 \end{bmatrix} \end{aligned} \tag{8.136}$$

In the remainder of this example, we use the symbol G to denote any matrix such that $GG^T = Q$. The differential Riccati equation for the Kalman filter is given as

$$\dot{P} = -PC^TR^{-1}CP + AP + PA^T + Q \tag{8.137}$$

This can be written as the following three coupled differential equations.

$$\begin{aligned}
\dot{p}_{11} &= 2a_1p_{11} - p_{11}^2/r_1 - p_{12}^2 + q_{11} \\
\dot{p}_{12} &= (a_1 + a_2)p_{12} - p_{11}p_{12}/r_1 - p_{12}p_{22}/r_2 + q_{12} \\
\dot{p}_{22} &= 2a_2p_{22} - p_{12}^2/r_1 - p_{22}^2/r_2 + q_{22}
\end{aligned} \tag{8.138}$$

We set these derivatives equal to zero to obtain the steady-state Riccati equation solution.

If $a_1 \neq a_2$ and $q_{12} \neq 0$, then (A, C) is detectable and (A, G) is stabilizable (see Problem 8.8). The results of Theorem 27 therefore apply to this situation. It can be shown that the unique positive semidefinite ARE solution in this case is

$$\begin{aligned}
p_{11} &= r_1 \left[a_1 + \left(\gamma_1 - \frac{p_{12}^2}{r_1 r_2} \right)^{1/2} \right] \\
p_{22} &= r_2 \left[a_2 + \left(\gamma_2 - \frac{p_{12}^2}{r_1 r_2} \right)^{1/2} \right] \\
p_{12} &= q_{12} \left[\gamma_1 + \gamma_2 + 2 \left(\gamma_1 \gamma_2 - q_{12}^2 / r_1 r_2 \right)^{1/2} \right]^{-1/2} \\
\gamma_1 &= \frac{q_{11}}{r_1} + a_1^2 \\
\gamma_2 &= \frac{q_{22}}{r_2} + a_2^2
\end{aligned} \tag{8.139}$$

This results in a stable steady-state Kalman filter.

If $a_1 = a_2 < 0$, $q_{12} \neq 0$, and $|Q| = 0$, then (A, C) is detectable, and (A, G) is stabilizable (see Problem 8.9). The results of Theorem 27 therefore apply to this situation as well. It can be shown that the unique positive semidefinite ARE solution in this case is given as

$$\begin{aligned}
p_{11} &= q_{11}/\gamma_3 \\
p_{22} &= q_{22}/\gamma_3 \\
p_{12} &= q_{12}/\gamma_3 \\
\gamma_3 &= -a_1 + (a_1^2 + q_{11}/r_1 + q_{22}/r_2)^{1/2}
\end{aligned} \tag{8.140}$$

This results in a stable steady-state Kalman filter.

If $a_1 = a_2 > 0$, $q_{12} \neq 0$, and $|Q| = 0$, then (A, C) is detectable and (A, G) is controllable on the imaginary axis, but (A, G) is not stabilizable (see Problem 8.10). The results of Theorem 27 do not apply to this situation, but Theorem 28 does apply to this situation. It can be shown that Equations (8.139) and (8.140) are both positive semidefinite ARE solutions in this case. If we integrate Equation (8.138) we may come up with Equation (8.139) as the steady-state solution, or we may come up with Equation (8.140) as the steady-state solution, depending on the initial condition $P(0)$. However, only one of the solutions will result in a stable Kalman filter.⁴

To be more specific, consider the case $a_1 = a_2 = 1$, $q_{11} = q_{12} = q_{22} = 0$, and $r_1 = r_2 = 1$. For these values, we can simulate the differential Riccati

⁴If we use MATLAB's CARE function then we will get the stabilizing solution.

equations of Equation (8.138) to find the steady-state Riccati solution, the steady-state Kalman gain, and the steady-state estimator, as follows:

$$\begin{aligned}
 P &= \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \text{ or } \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \\
 K &= PC^T R^{-1} \\
 &= \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \text{ or } \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \\
 \dot{\hat{x}} &= (A - KC)\hat{x} + Ky \\
 &= (-\hat{x} + Ky) \text{ or } (\hat{x} + Ky)
 \end{aligned} \tag{8.141}$$

The ARE solution depends on the initial condition $P(0)$. The first ARE solution results in a positive semidefinite ARE solution that gives a stable Kalman filter. The second ARE solution results in a positive semidefinite ARE solution that gives an unstable Kalman filter. This agrees with Theorem 28.

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8.5.2 The Wiener filter is a Kalman filter

Consider the steady-state continuous-time Kalman filter.

$$\dot{\hat{x}} = Ax + K(y - C\hat{x}) \tag{8.142}$$

Taking the Laplace transform of both sides of this equation gives

$$\begin{aligned}
 (sI - A + KC)\hat{X}(s) &= KY(s) \\
 \hat{X}(s) &= (sI - A + KC)^{-1}KY(s)
 \end{aligned} \tag{8.143}$$

The transfer function from $y(t)$ to $\hat{x}(t)$ is identical to the transfer function of the Wiener filter [Buc68, Chapter 5], Sha82, [Sag71, Chapter 7]. In other words, the Wiener filter is a special case of the Kalman filter. The equivalence of discrete-time Wiener and Kalman filtering is discussed in [Men87].

■ EXAMPLE 8.6

Consider the scalar system given by

$$\begin{aligned}
 \dot{x} &= -x + w \\
 y &= x + v
 \end{aligned} \tag{8.144}$$

where w and v are zero-mean, uncorrelated white noise processes with respective variances $Q = 2$ and $R = 1$. The steady-state Kalman filter for this system can be obtained by solving Equation (8.37) with $\dot{P} = 0$, from which we obtain

$$\dot{\hat{x}} = -\sqrt{3}\hat{x} + (\sqrt{3} - 1)y \tag{8.145}$$

Taking the Laplace transform of this estimator gives

$$(s + \sqrt{3})\hat{X}(s) = (\sqrt{3} - 1)Y(s) \tag{8.146}$$

In other words, the Kalman filter is equivalent to passing the measurement $y(t)$ through the transfer function $G(s)$, which is given as

$$G(s) = \frac{\sqrt{3} - 1}{s + \sqrt{3}} \quad (8.147)$$

The impulse response of the Kalman filter is obtained by taking the inverse Laplace transform, which gives

$$g(t) = (\sqrt{3} - 1)e^{-\sqrt{3}t}, \quad t \geq 0 \quad (8.148)$$

Now we will obtain the power spectrum of the state by taking the Laplace transform of Equation (8.144). This gives

$$\begin{aligned} sX(s) &= -X(s) + W(s) \\ X(s) &= \frac{1}{s+1}W(s) \end{aligned} \quad (8.149)$$

We see that the state $x(t)$ can be obtained by passing the white noise $w(t)$ (which has a power spectrum $S_w(\omega) = Q = 2$) through the transfer function $L(s) = 1/(s+1)$. From Equation (3.75) we see how to compute the power spectrum of the output of a linear system. This gives the power spectrum of $x(t)$ as

$$\begin{aligned} S_x(\omega) &= L(-\omega)L(\omega)S_w(\omega) \\ &= \left(\frac{1}{-j\omega+1}\right)\left(\frac{1}{j\omega+1}\right)2 \\ &= \frac{2}{\omega^2+1} \end{aligned} \quad (8.150)$$

The causal Wiener filter for a signal with this power spectrum, corrupted by white measurement noise with a unity power spectrum, was obtained in Example 3.10. The Wiener filter was found to be identical to the steady-state Kalman filter of Equation (8.148). This example serves to illustrate the equivalence of Wiener filtering and steady-state Kalman filtering.

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8.5.3 Duality

It is interesting to note the duality between optimal estimation and optimal control. The optimal estimation problem begins with the system and measurement equations

$$\begin{aligned} \dot{x} &= Ax + w \\ w &\sim N(0, Q) \\ y &= Cx + v \\ v &\sim N(0, R) \end{aligned} \quad (8.151)$$

Recall that Q and R are symmetric matrices. The optimal estimation problem tries to find the state estimate $\hat{x}$ that minimizes the cost function

$$J_e = \int_0^{t_f} E[(x - \hat{x})^T(x - \hat{x})] dt \quad (8.152)$$